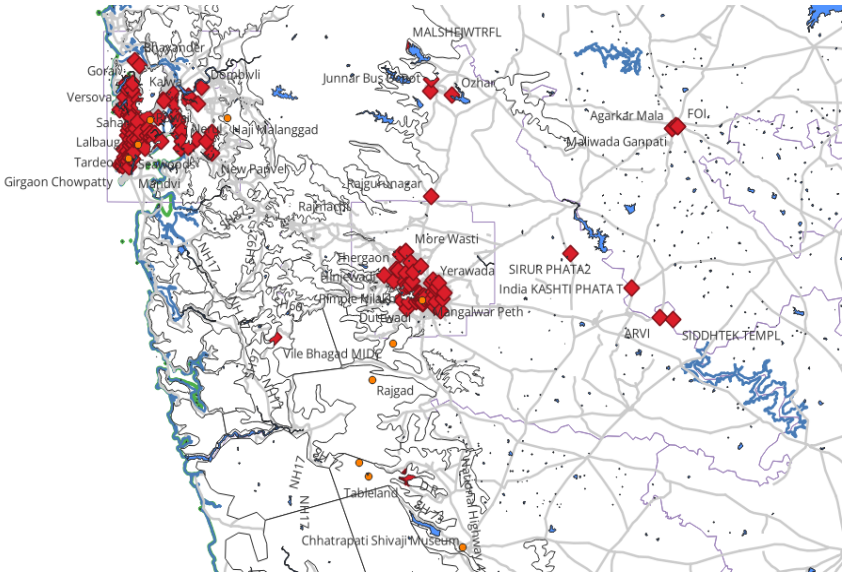
Practical 4

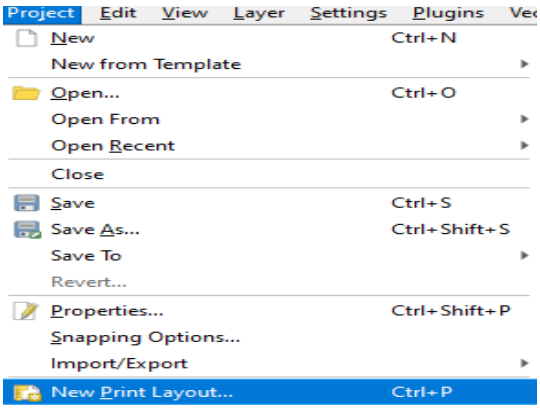
Making a Map, Working with attributes, Importing Spreadsheets or CSV files using plugins, searching and downloading OpenStreetMap Data

A. Making a Map

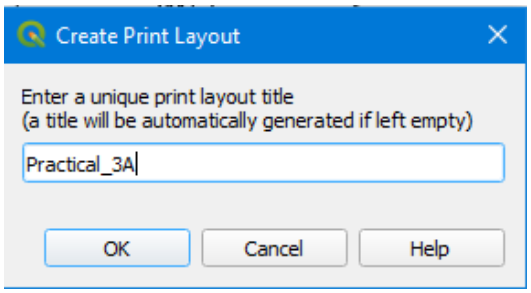
- Create a new thematic map or open and existing one.
- Consider the following map as an example map.



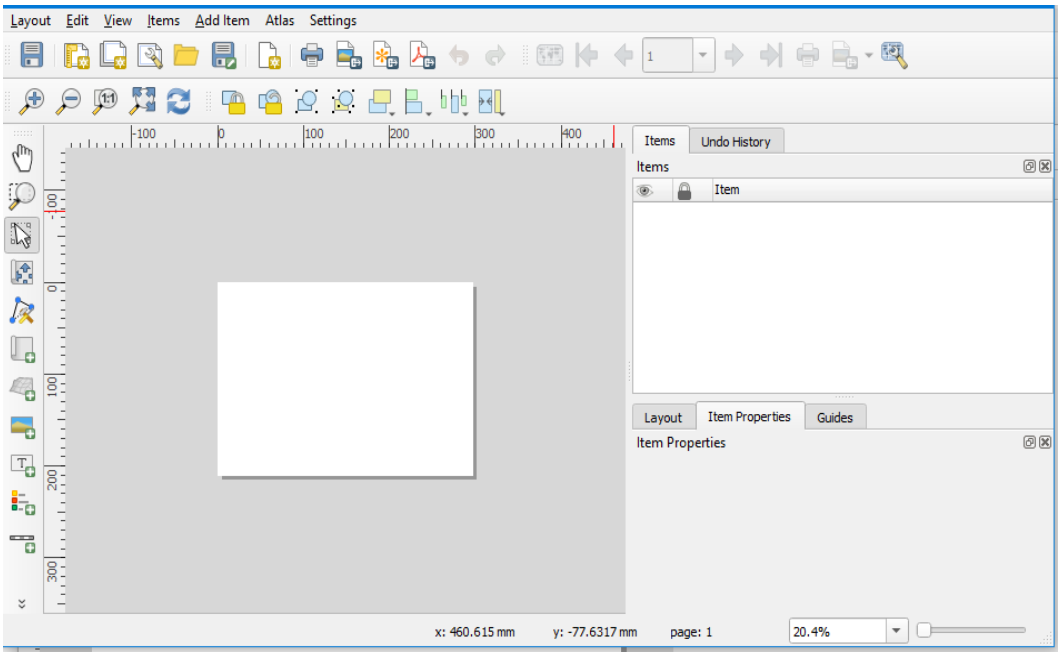
- Go to Protect→New Print Layout.



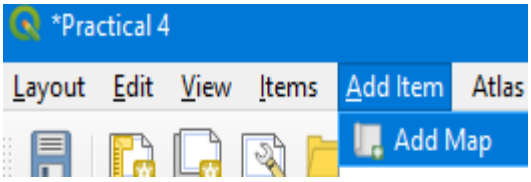
- Insert a suitable title and press "OK".

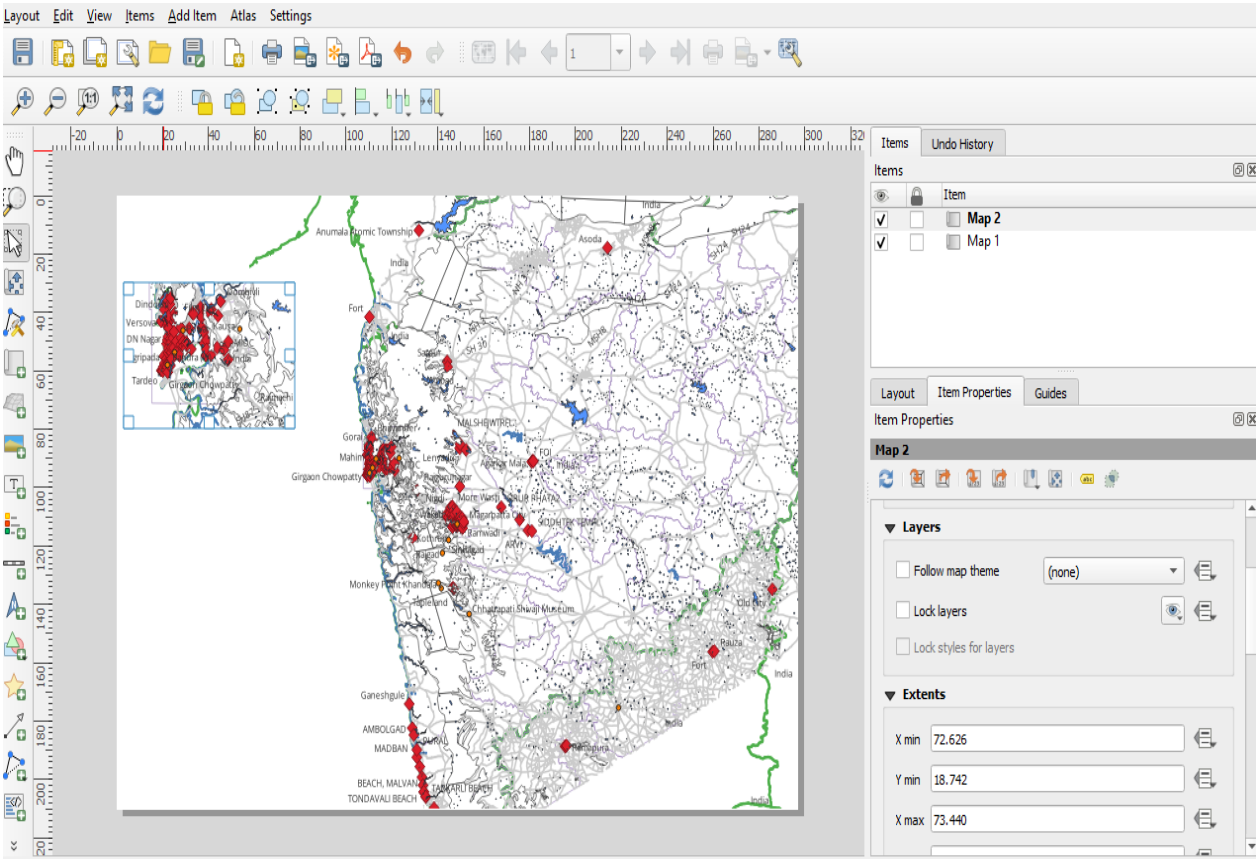


- A new Print Layout window will open.

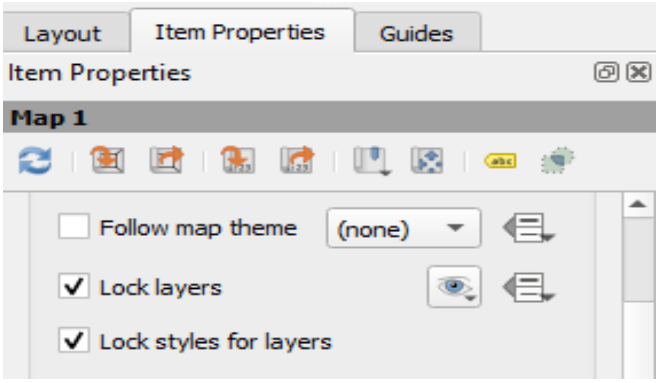


- Select Add item→Add Map.



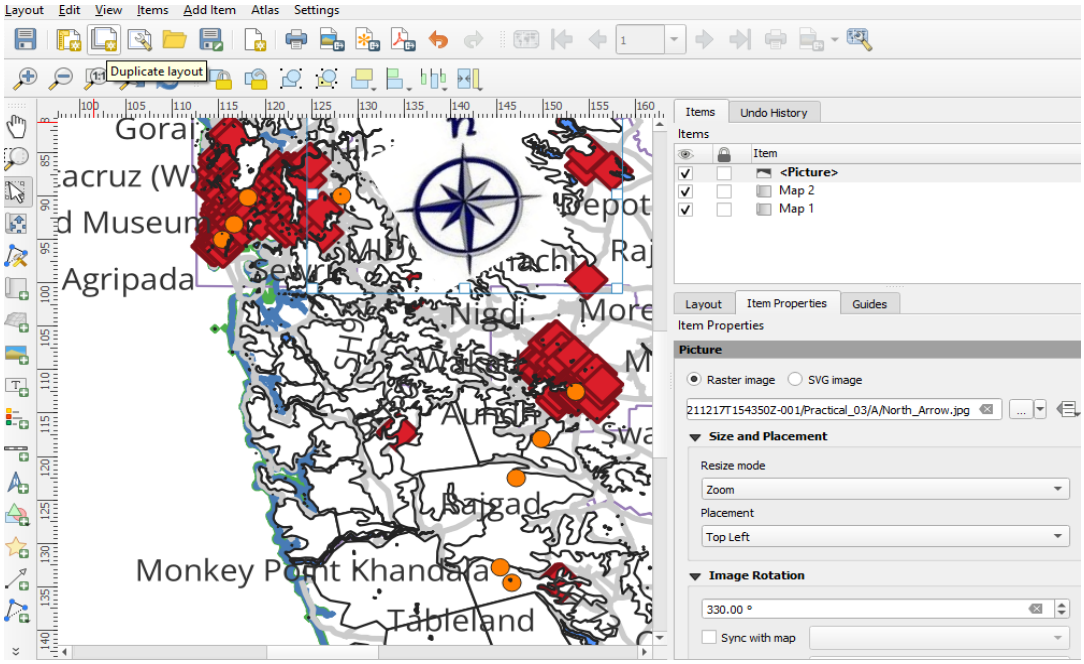


- After adding map go to Item Properties→Map1→Layers.
- Check on Lock Layers and Lock Styles for Layers.

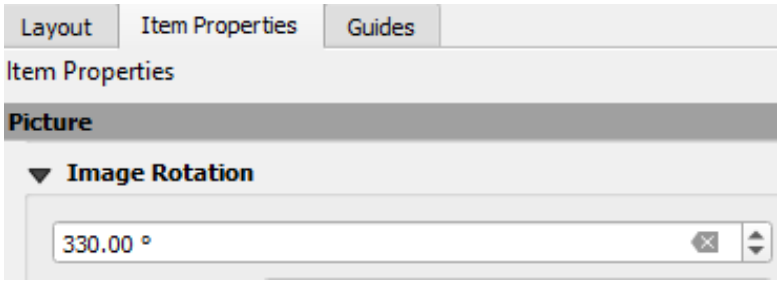


This will ensure that if any change in layers or change their styles, the print layout view not change.

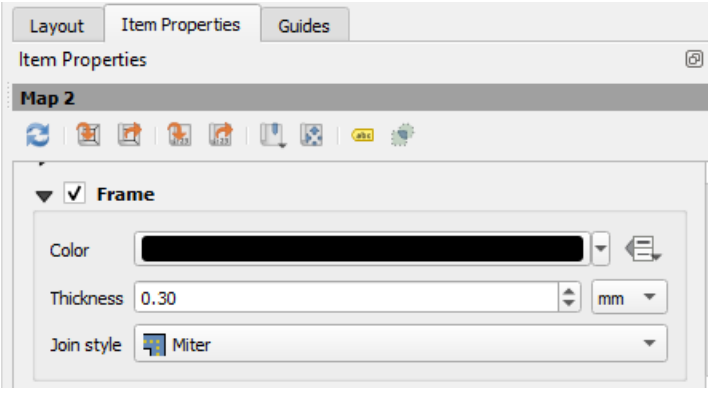
- Go to Add Item→Add Picture→ Place a picture box at appropriate location.



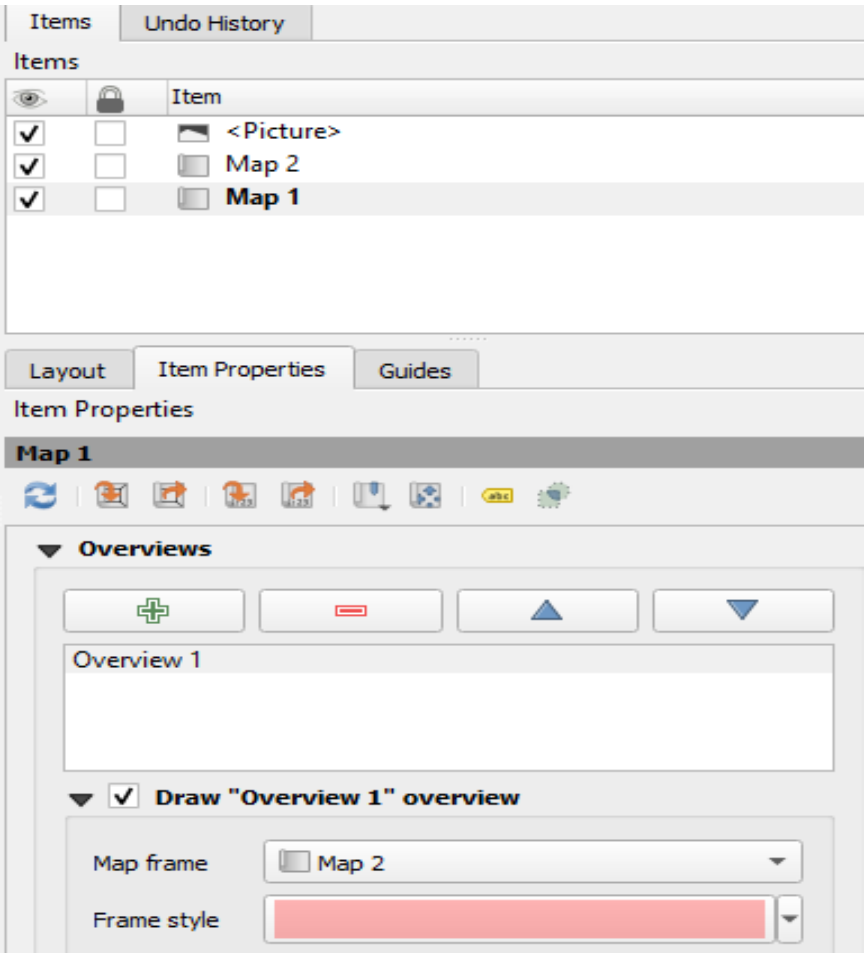
- Also adjust Image Rotations to its appropriate value.
- Item Properties →Image Rotation



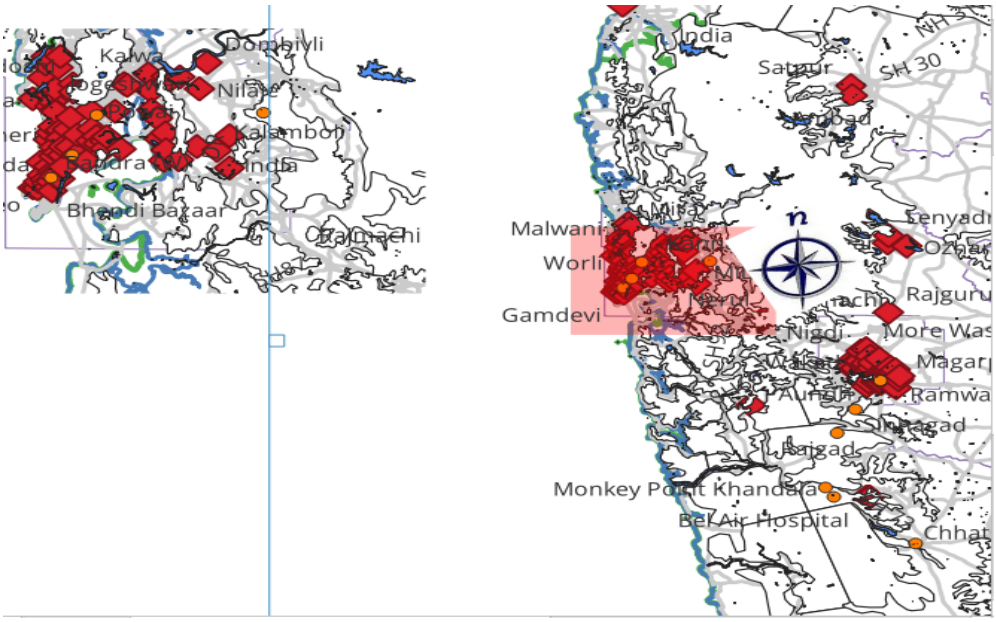
- Add an inset Using Add Item → Add Picture→ Select an area to be highlighted on Main Map.
- Set a frame for inset by enabling the check box for Frame.



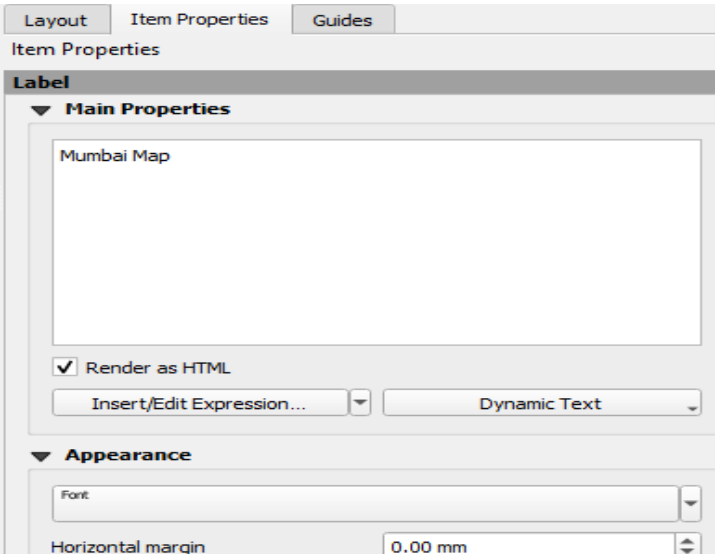
- To highlight the area shown in Inset.
- Select the Picture representing main Map from Items Pane.
- In Item Properties →Overviews→ Using icon add an overview.
- Select the checkbox Draw Overview.
- Name the picture object representing inset (Map1 in our case.)



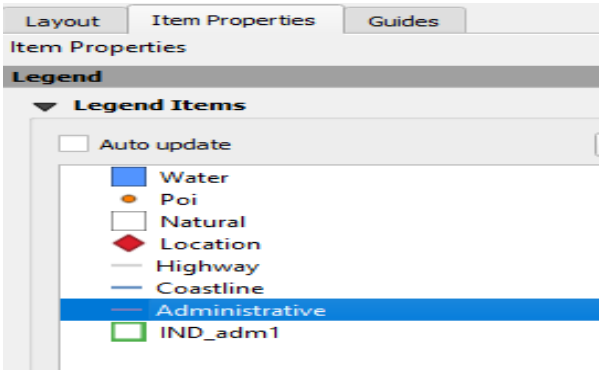
- The Print Layout will appear like.



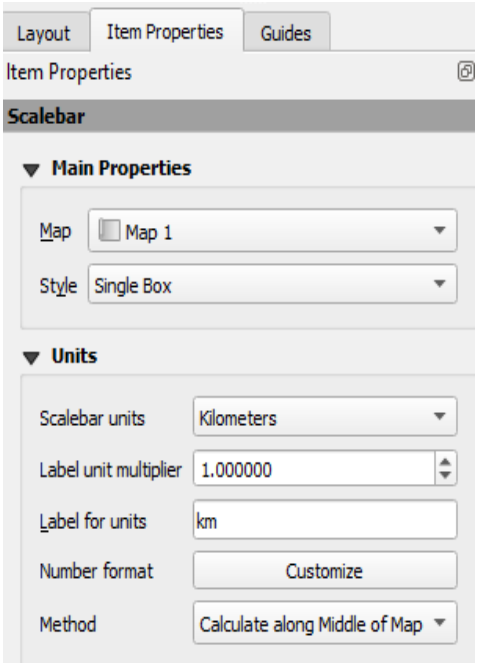
- Add Item→Add Label.
- Change the Label text to “Mumbai Map”, Set appropriate font size & color using Item.
- Item Properties→Main Properties.

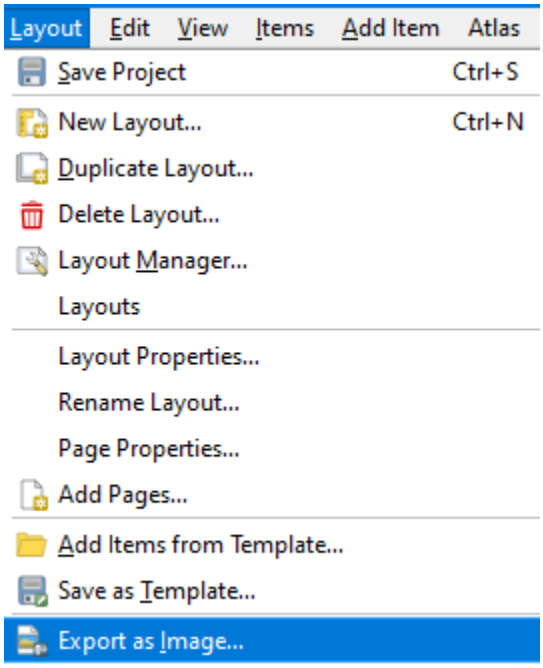


- Add Item→Add Legend→Place the legend indicator at appropriate location.
- Uncheck Auto Update and use suitable legend indicator label.

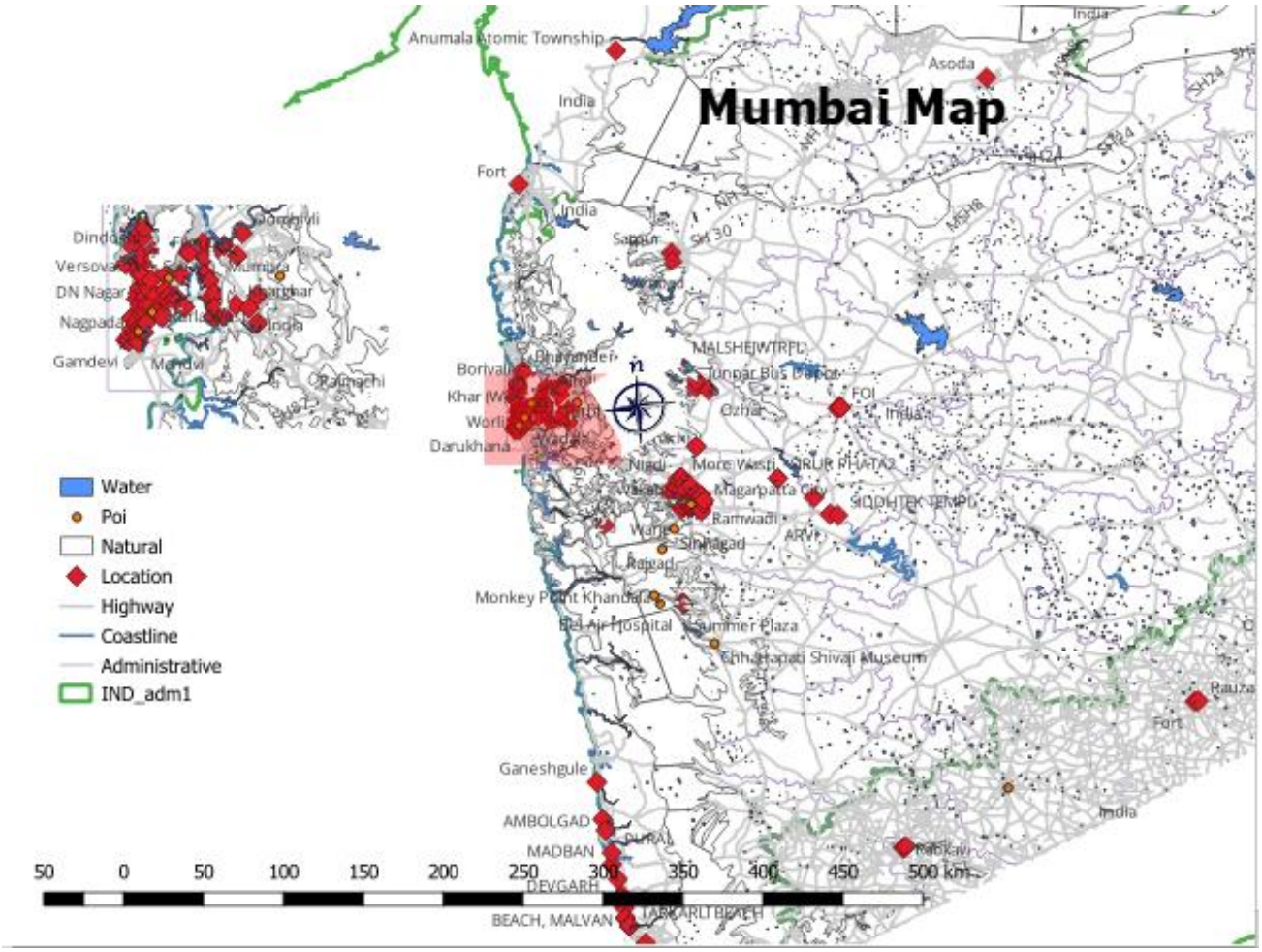


- Add Item→Add Scale Bar
- A Map can be saved Image or PDF using Layout→Export as Image/Export as PDF.



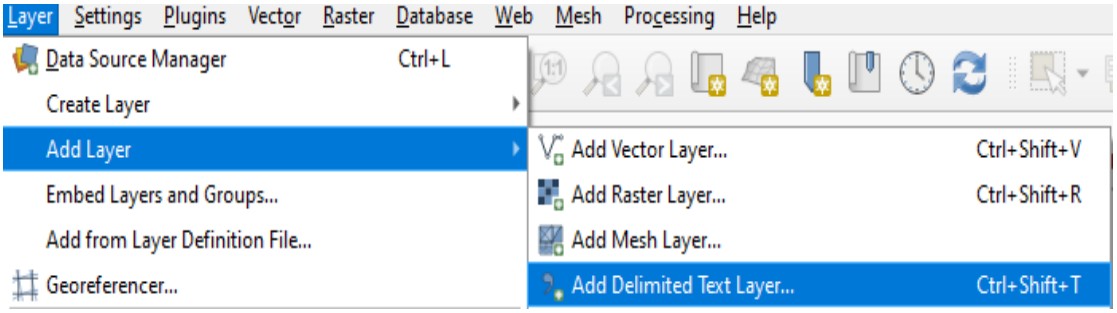


- Save the Map to a location appropriate location as PDF or Image.

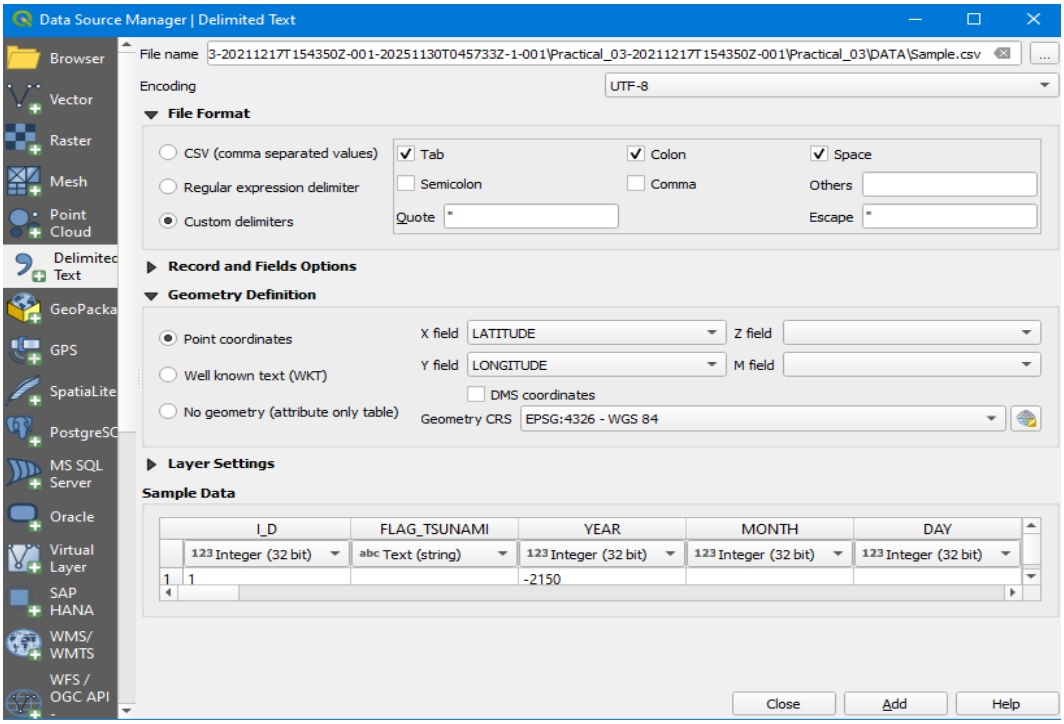


B. Importing Spreadsheets or CSV Files

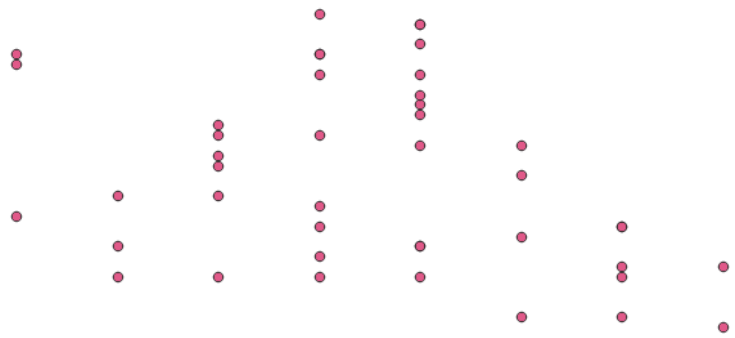
- Many times the GIS data comes in a Table or an Excel spreadsheet or a list lat/ long coordinates. Therefore it has to be imported in a GIS project.
- Sample file for Earthquake data will be used in this practical.
- Go to Layer →Add Layer→Add Delimited text Layer.



- Data Source Manager | Delimited Text window will appear.
- Select the \GIS_Workshop\Practicals\Practical_03\C\Sample.csv file from data folder.

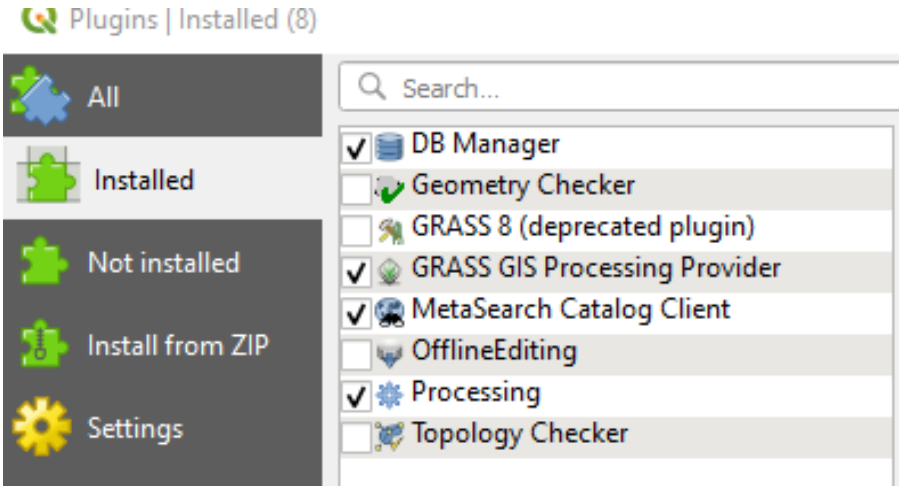


- Press ADD and close the window.
- Output:

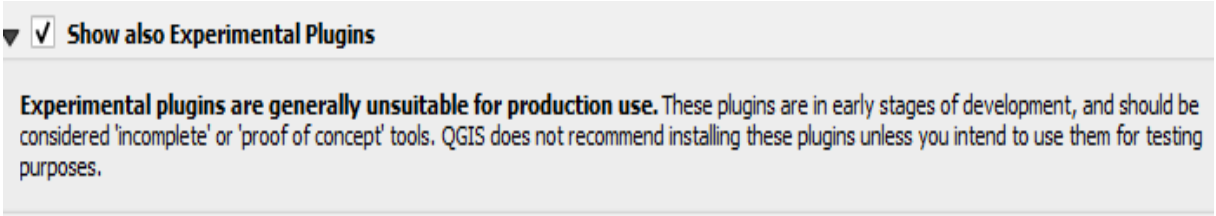


C. Using Plugins

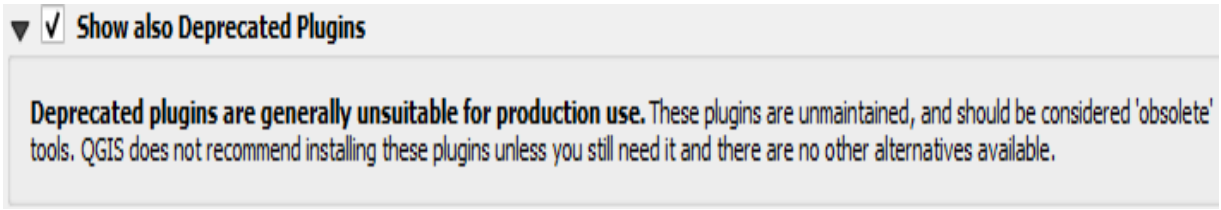
- Core plugins are already part of the standard QGIS installation to use these, just enable them.
- Open QGIS. Click on plugins→Manage and Install Plugins.

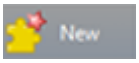


- **To enable a plugins**, check on the checkbox next to plugin. This will enable the plugin to use it.
- **External plugins** are available in the QGIS Plugins Repository and need to be installed by the users before using them.
- Click on Not Installed or Install from ZIP.
- Once the plugin is downloaded and installed, you will see a confirmation dialog.
- Click on Plugins→<<new Plugin Name>>
- The plugin if marked **Experimental plugin** can be installed, from setting→check on.



OR



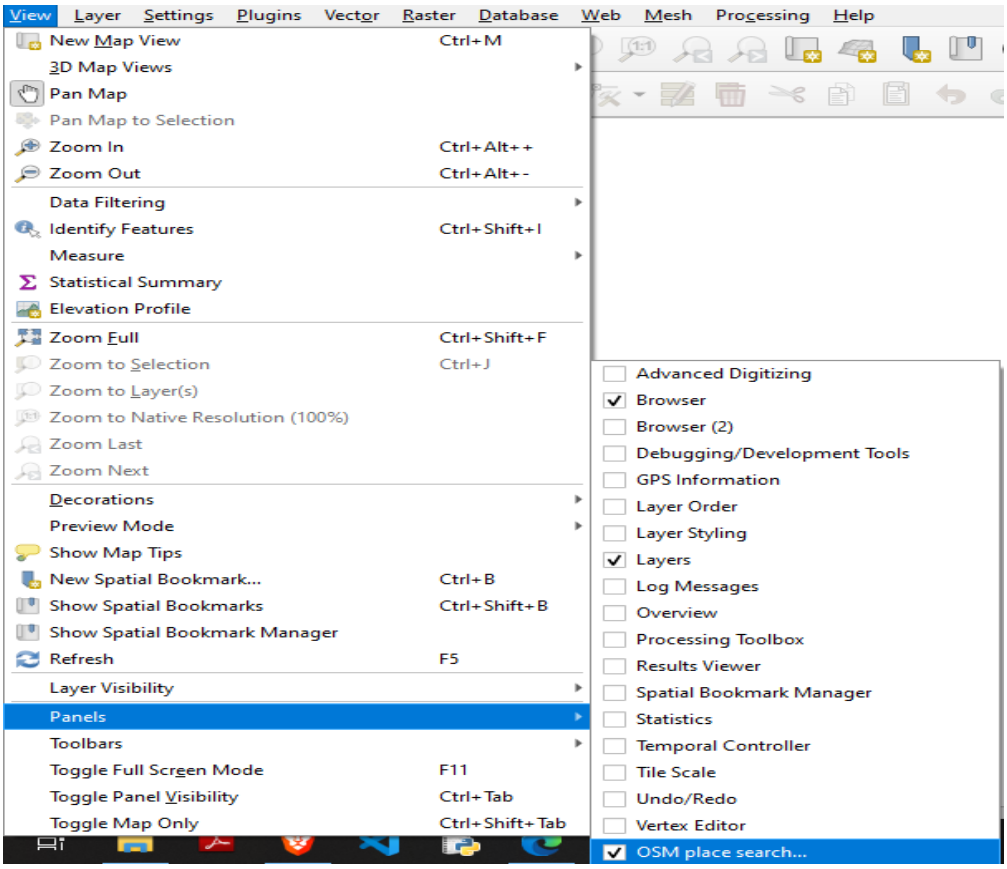
- A 

tab will be added to Plugin Manager Window.
- Click on a plugin name and Click Install.

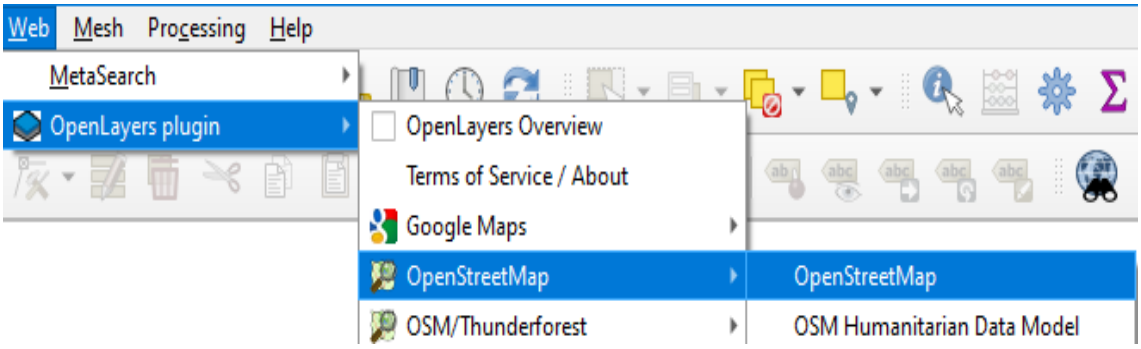
D. Searching and Downloading OpenStreetMap Data

OpenStreetMap (OSM) created by Steve Coast in the UK in 2004 is a collaborative project to create a free editable map of the world. Rather than the map itself, the data generated by the project is considered its primary output. The creation and growth of OSM has been motivated by restrictions on use or availability of map information across much of the world and the advent inexpensive portable satellite navigation devices.

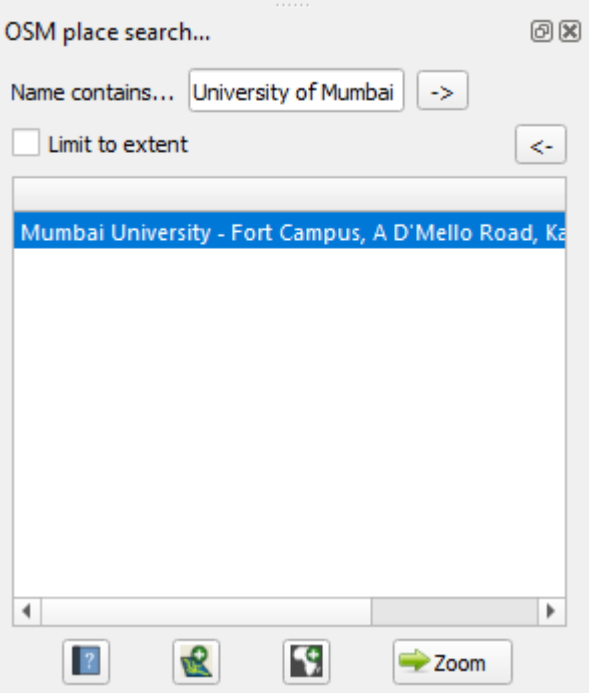
- Add **"Open Layer"** and **"OSM Search"** Plugin from Not Installed option from Plugin Manager Dialog Box.
- The OSM Place Search plugin will install itself as a Panel in QGIS, if not to go to view →Panels→Select OSM Place Search.

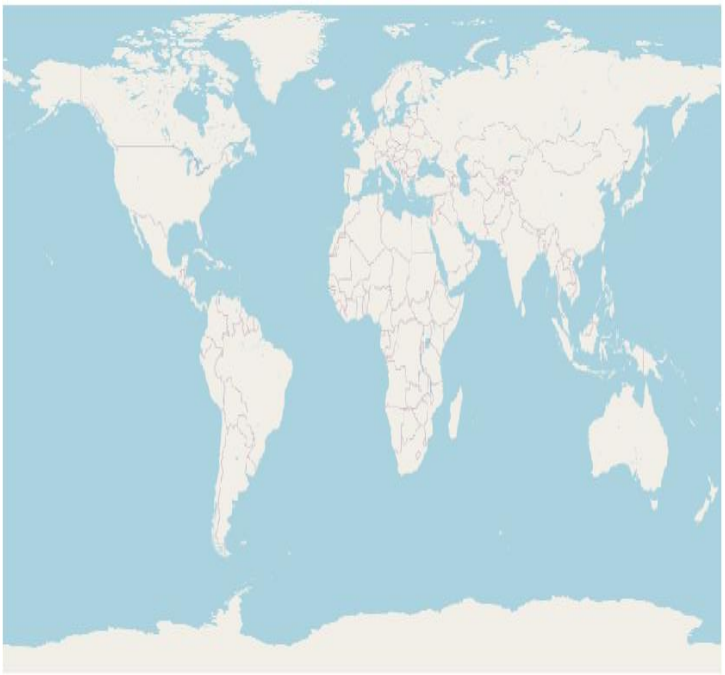


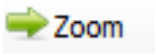
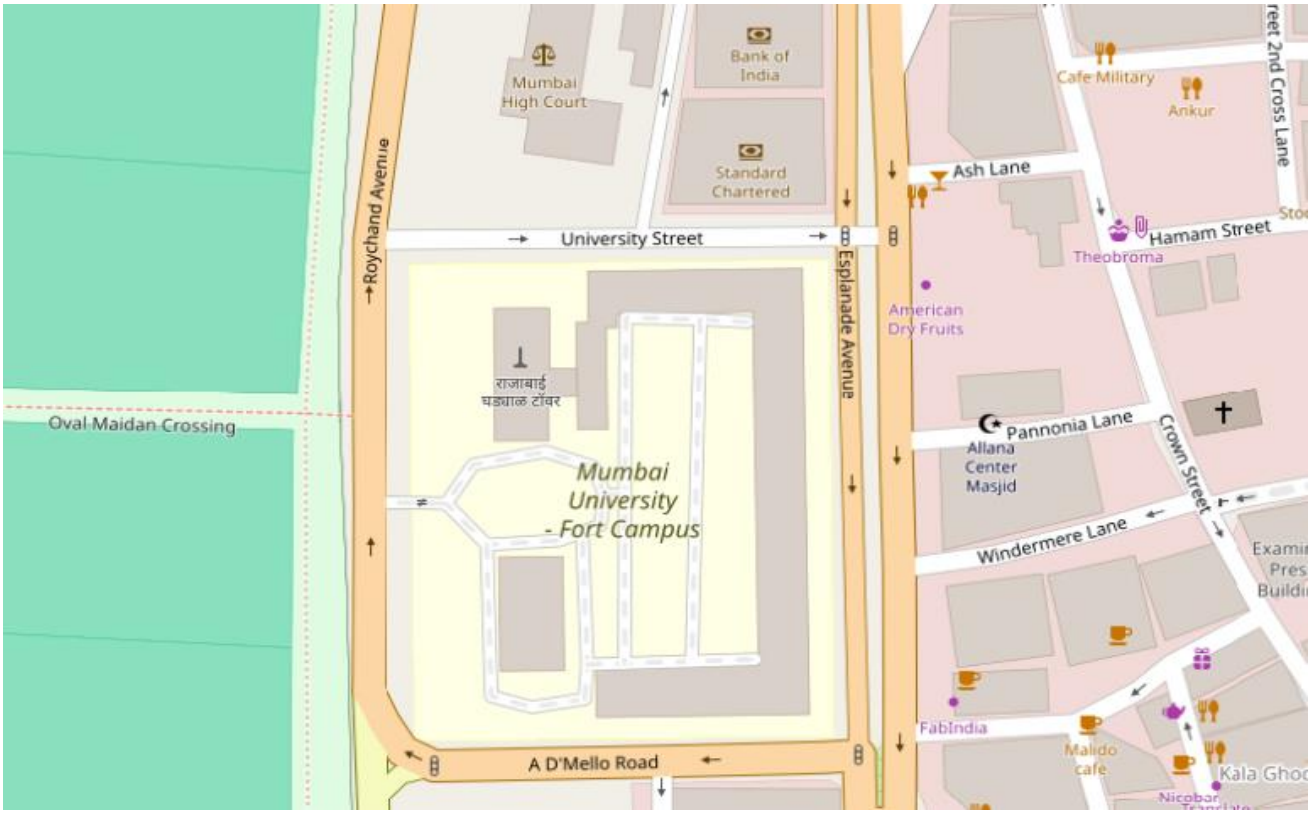
- Go to Web→OpenLayers Plugin and Select Open Street Map.



- A World map will appear on screen.
- If an error occurs in loading maps, go to project properties →CRS→**Project Coordinate Reference System (CRS).**





- In OSM Place search Pane→Enter Mumbai or any place name to search.
- Double click on the desired place in OSM Place search Panel or Click and press 

Output: